INCH-POUND

MIL-F-8839A
23 December 1988
SUPERSEDING
MIL-F-8839(ASG)
23 July 1958

MILITARY SPECIFICATION

FITTINGS, LUBRICATION, DRIVE TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers drive-type lubrication fittings for maintenance use only and is not intended for new design.

1.2 Classification. Lubrication fittings covered by this specification shall be of the following classes and types.

1.2.1 Type.

Type I - Plain end.

Type II - Serrated end.

Type III - Flush end.

1.2.2 Classes.

Class 1 - Drive type fitting with no check valve.

Class 2 - Drive type fitting with check valve.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and publications. Unless otherwise specified, the following specifications, standards and publications of the issue listed in that issue of the Department of Defense, Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Systems Engineering and Standardization Department (Code 53), Naval Air Engineering Center, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

QQ-N-281	Nickel-Copper-Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings and Structural and Special Shaped Sections
QQ-W-390	Wire, Nickel-Chromium Iron Alloy
QQ-P-416	Plating, Cadmium (Electrodeposited)
PPP-B-566	Boxes, Folding, Paperboard
PPP-B-576	Boxes, Wood, Cleated, Veneer, Paper Overlaid
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-591	Boxes, Shipping, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-B-665	Boxes, Paperboard, Metal Stayed (Including Stay Material)
PPP-B-676	Boxes, Set-Up, Paperboard
PPP-T-60	Tape, Packaging, Waterproof

MILITARY

MIL-P-116	Preservation, Methods of
MIL-F-3541	Fittings, Lubrication
MIL-L-10547	Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible

STANDARDS

MILITARY

DOD-STD-100	Engineering Drawing Practices
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

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STANDARDS (Continued)

MILITARY (Continued)

MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-2073-1	DOD Material Procedures for Development and Application of Packaging Requirements
MS15000	Fittings, Lubrication-Hydraulic, Acceptable Tip Designs

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Naval Publications and Forms Center (Attn: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Nongovernment publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue listed in the current DODISS and supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A108	Steel Bars Carbon Cold-Finished, Standard Quality Specification for
B633	Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel
D3951	Packaging, Commercial
E18	Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, Test Methods for

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Classification of requirements. The requirements for the product are classified as follows:

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Requirements	Paragraph
Materials	3.2
Design and construction	3.3
Protective coating	3.5
Performance	3.6
Part numbering of interchangeable parts	3.7
Dimensions and tolerances	3.8
Identification of product	3.9
Workmanship	3.10

3.2 Materials.

3.2.1 Balls and tips. Steel per ASTM A108 or Monel per QQ-N-281.

3.2.2 Springs. Nickel-chromium-iron alloy per QQ-W-390 Condition C.

3.2.3 Finish. Steel-cadmium plated in accordance with QQ-P-416 Type II, Class 2. Monel - None.

3.3 Design and construction.

3.3.1 Drive type fittings. Drive type lubrication fittings shall conform to the shapes and dimensions shown on the applicable specification sheets.

3.3.1.1 Drive shanks. The drive shanks on the lubrication fittings shall be suitable for installing the fitting by driving it into a drilled hole.

3.3.2 Body. All lubrication fittings shall be fabricated from steel per ASTM A108.

3.3.3 Check valves. Unless otherwise specified, lubrication fittings shall be provided with a check valve which will readily admit lubricant but will prevent its escape through the inlet tip. The valve shall be so constructed that leakage will not occur when tested in accordance with 3.5.1.

3.3.4 Inlet tip. The inlet tip of the lubrication fittings shall conform to the latest revision of MS15000 design standard. The surface of the carbon steel tip that contacts with the nozzle or coupling of the gun shall be case-hardened to a depth of 0.005 to 0.009 inch with a minimum reading of 83 on the Rockwell 15-N scale. The tip of R-monel fitting shall be heat treated to a minimum hardness of 270 Brinell or hardness no. 27 on Rockwell "C" scale. The tip shall be integral with the body of the lubrication fitting or securely attached by means of screw threads.

3.4 Protective coating. The external surfaces of the steel fittings shall be cadmium plated in accordance with specification QQ-P-416, Type II, Class 3, or zinc plated in accordance with specification ASTM B633.

3.5 Performance. The lubrication fittings shall be suitably connected with gun couplers and shall admit lubricant at 1000 psi without leakage between the gun couplers and the fittings, when the gun nozzle is connected to

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the fitting at a minimum angle of 10 degrees in any radial direction between the axis of the gun, nozzle or the coupler, and the axis of the inlet tip of the fitting as shown in the design standard.

3.5.1 Lubricant leakage. All lubrication fittings equipped with a check valve shall retain soap-type grease lubricant at a pressure of 1800 psi without leakage using a test gage block (see Figure 1).

3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of DOD-STD-1000 shall govern the manufacturer's part numbers and changes thereto.

3.7 Dimensions and tolerances. All pertinent dimensions and tolerances, where interchangeability, operation or performance of the fitting may be affected, shall be as specified on the manufacturer's drawings.

3.8 Identification of product. Each lubrication fitting shall be marked for identification in accordance with standard MIL-STD-130 and the manufacturer's name or trademark shall be permanently marked on all lubrication fittings.

3.9 Workmanship. Remove all burrs and break all sharp edges. The outside tip surfaces shall be free of detrimental longitudinal and spiral tool marks. All fitting outside surfaces shall be smooth, free from irregularities and defects which could adversely affect safety, performance, reliability or durability.

3.10 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements. Unless otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All lubrication fittings must meet all the requirements of Sections 3 and 5 as applicable. The inspection set forth shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, and does not commit the Government to accept defective material.

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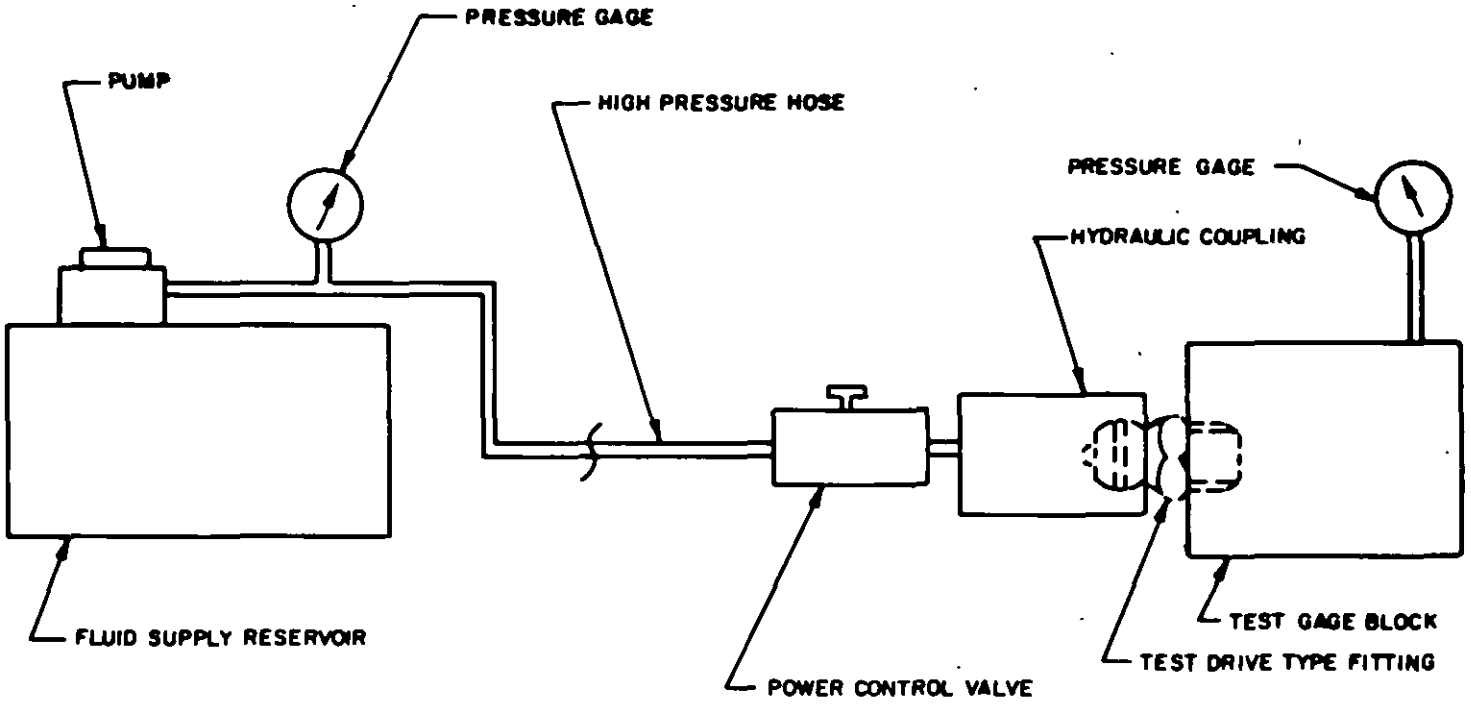


FIGURE 1. Leakage test setup.

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4.2 Classification of inspection. All of the inspections of fittings shall be classified as quality conformance inspection.

4.2.1 Quality conformance inspection. Quality conformance inspection shall consist of sampling inspection only.

4.2.2 Sampling inspection. A random sample shall be selected from each inspection lot and subjected to the following inspections.

4.2.2.1 Inspection lot. A lot shall consist of lubrication fittings of the same size, part number, made from the same batch of material of the same contract or order, submitted for inspection at the same time.

4.2.2.2 Record maintenance. The supplier shall maintain a record of material certification, chemical analysis and inspection records and the inspection records shall be available upon request by the contracting agency.

4.2.2.3 Material certification. Records of chemical analysis and mechanical property tests showing conformance to the applicable material specifications shall be retained by the material supplier and shall be available upon request by the contracting agency.

4.2.2.4 Resubmitted inspection lots. A resubmitted inspection lot shall be inspected by the contractor under supervision of the Government Inspector using tightened inspection. If the original acceptance number were zero, a sample size represented by the next higher sample size code letter shall be chosen with the acceptance number remaining zero. Before an inspection lot is resubmitted, full particulars concerning the cause of previous rejection, and the action taken to correct the defects found in the inspection lot, shall be furnished by the contractor to the Government Inspector.

4.3 Inspection methods.

4.3.1 Examination of product. The fitting shall be thoroughly examined to determine conformance with this specification and applicable drawings with respect to all requirements not covered by tests specified herein.

4.3.2 Examination of preparing for delivery. The preservation packaging, packing and marking of the fittings shall be examined to conform with the applicable requirements of Section 5 identified to referenced specifications and standards shall be inspected in accordance with the applicable requirements of these documents.

4.3.3 Operation.

4.3.3.1 Opening pressure. The fitting shall be tested utilizing a suitable test setup. A maximum pressure of 450 psi shall be applied to the inlet tip until the check valve opens and lubricant flows through the tip. This pressure test is applied only to the Class 1 lubrication fitting.

4.3.3.2 Extreme angular position. The lubrication fitting shall be assembled into a hand gun and coupling. The longitudinal axis of the hand gun and coupling shall be displaced 15°, + or -0 degrees from the longitudinal

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axis of the fitting. The fitting shall admit lubricant. This test shall be repeated with the hand gun and coupling realigned to the opposite angular position of 15°, +1 or -0 degrees displacement.

4.3.4 Installation test.

4.3.4.1 Firmness. The lubrication fitting equipped with check valve after installation shall be tested for firmness with maximum pressure of 2000 psi applied to the fitting. The fitting shall remain seated firmly without any movement.

4.3.4.2 Lubricant leakage. The fitting shall be tested in a suitable test equipment. Lubricant shall be applied into the fitting with maximum pressure of 1800 psi. No lubricant leakage shall be found when the coupling disconnected from the fitting tip (see 3.5.1).

4.3.5 Hardness test. The lubrication fitting shall be tested for hardness in accordance with ASTM E18. The hardness shall conform to the values specified in 3.3.4.

4.3.6 Inspection of packaging. The sampling and inspection of the preservation, packing and container marking shall be in accordance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be the following level A or commercial as specified in the contract (see 6.2).

5.1.1 Level A. Unless otherwise specified, fittings in a unit quantity of 100 each, of the same part number, shall be preserved in accordance with Method III of specification MIL-P-116. Fittings shall be packaged in unit containers conforming to PPP-B-566 or PPP-B-676.

5.1.2 Commercial. The fittings shall be packaged in accordance with ASTM D-3951.

5.2 Intermediate packaging. When the quantity permits, five unit packages or a multiple thereof shall be further packaged in intermediate containers conforming to specification PPP-B-566, PPP-B-676, or PPP-B-665. The gross weight of the intermediate containers shall not exceed 20 pounds.

5.3 Packing. Packing shall be level A, B or commercial, as specified in the contract or order (see 6.2).

5.3.1 Level A. Fittings shall be packed in overseas-type boxes conforming to any of the following specifications: PPP-B-585, Class 3, PPP-B-591, PPP-B-601, PPP-B-621, (see 6.2) or PPP-B-576. Gross weight of wood and wood-cleated boxes shall not exceed 200 pounds. Box closures shall be in accordance with the box specification. Boxes shall be provided with case liners conforming to specification MIL-L-10547, type 1, Grade A.

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5.3.2 Level B. Fittings shall be packed for shipment in exterior shipping containers (domestic type) conforming to specifications PPP-B-585, Class 1, PPP-B-601 (domestic type), PPP-B-621, Class 1, PPP-B-636, Type CF or SF weather resistant, or PPP-B-576 (domestic type). Closures shall be in accordance with the applicable box specification. The gross weight of the shipping container when packed for shipment shall not exceed 200 pounds. The gross weight of fiberboard boxes shall not exceed the weight limitation of the applicable box specification.

5.3.3 Level C. The level C packing for fittings shall conform to MIL-STD-2073-1.

5.4 Marking of shipments. Interior and exterior shipping containers shall be marked in accordance with standard MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The fittings covered by this specification are intended for maintenance use on aircraft, ground vehicles, submarines, surface vessels, and industrial equipment where suitable, to facilitate necessary lubrication. For applications involving use of threaded fittings, specification MIL-F-3541 should be used.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Type of fitting, MS part number and manufacturer part number.

c. Quantity per shipping container when required.

d. Levels of preservation and packaging and packing, and marking of shipments required (see 5.1 and 5.3).

e. Data items required (see 6.3).

6.3 Data. For the information of contractors and contracting agency, any of the data specified in (a) applicable documents listed in Section 2 of this specification, or (b) referenced lower-tier documents need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order. The data to be furnished shall be listed on DD Form 1423 (Contract Data Requirements List), which shall be attached to and made a part of the contract or order.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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6.5 Subject term (keyword) listing.

Fittings, lubrication

Drive type

Maintenance use

Aircraft, ground vehicles, submarines, surface vessels and industrial equipment

Custodians:
Army - AV
Navy - AS
Air Force - 99

Review activities:
Army - AR, MI, AT
Air Force - 82
Navy - SH, OS

User activities:
Army - CE
Navy - MC, OS, SH

Preparing Activity
Navy - AS
(Project No. 4730-1067)

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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